Alexandru Balta

Cluj-Napoca, Romania | alexandrubalta222@gmail.com

linkedin.com/in/alexandru-balta | github.com/kerBiy | kerbiy.github.io/web-portfolio

Professional Summary

Full-stack software engineer (Java & React) and MSc student in Distributed Systems. Shipped a production feature at 8x8 (Spring Boot, React) and built Crate, an event-driven microservices platform with Spring Boot, Kafka and PostgreSQL. Available for full-time backend or full-stack roles, on-site or hybrid in Cluj-Napoca, or remote across Europe.

Education

Master's Degree in Distributed Systems — Oct 2026 – Jul 2028 (expected)
Babes-Bolyai University — *Cluj-Napoca, Romania*

Bachelor's Degree in Computer Science — Oct 2023 – Jul 2026
Babes-Bolyai University — *Cluj-Napoca, Romania*

Work Experience

Software Engineer Intern — Jul 2025 – Sep 2025
8x8 — *Cluj-Napoca, Romania*

- Built and shipped to production the **Favorite Customers** feature (Java Spring Boot, Hibernate/JPA, React, Liquibase) for an internal tools platform used daily by **100+ employees**, where client search is a core workflow.
- Engineered an asynchronous mechanism that preloads and caches favorite client profiles in the background, replacing a slow ∼1s client lookup with one-click access from cache.
- Implemented global state management with **Redux Toolkit**, including optimistic UI updates that render user actions immediately while requests resolve in the background.
- Delivered the feature in an **Agile/Scrum** team (**Jira**) with unit and integration tests (**JUnit**, **Mockito**) and peer code reviews, then presented it to engineering executives and stakeholders in California.

Projects

Crate — **Letterboxd for Albums** | *Java, Spring Boot, Kafka, PostgreSQL, React, TypeScript, Docker*

- Designed a platform for rating and reviewing albums with friends: a **React** frontend, a **Spring Cloud Gateway** with JWT auth, and three **Spring Boot** microservices with isolated **PostgreSQL** databases.
- Synchronized rating aggregates across services with **Kafka** events and idempotent consumers, verified by 149 **Testcontainers** integration tests (262 total) in **GitHub Actions**.
- Integrated the MusicBrainz API under its 1 req/s limit with a custom token-bucket rate limiter and a read-through cache, cutting repeat searches from ∼1.7s to ∼12ms.
- Created a search evaluation harness and used it to redesign ranking with ListenBrainz popularity data, raising top-3 accuracy from **13/30 to 28/30** and reaching 24/25 on a held-out query set.

SNN Brain Explorer | *React, FastAPI, Python, C++, OpenAI API*

- Developed a visualization tool for latency-coded Spiking Neural Networks (SNNs) as a Bachelor's Thesis with the **CRIStAL Laboratory (Univ. of Lille)**, making neuron timing visible where other tools only showed outputs.
- Built a FastAPI backend that turns canvas drawings into MNIST inputs for the lab's C++ simulator and a **scikit-learn** classifier, behind a React dashboard with a timeline slider to replay neuron activity layer by layer.
- Integrated a RAG-powered GPT-4o assistant that explains SNN dynamics and diagnoses classification errors.

Technical Skills

Programming Languages: Java, TypeScript, JavaScript, SQL, Python, C++, Bash, HTML/CSS
Frameworks & Libraries: Spring Boot, Hibernate/JPA, React, Redux Toolkit, Node.js, FastAPI, Tailwind
Data & Messaging: PostgreSQL, Kafka, MongoDB, Liquibase, RESTful APIs, JWT
Testing & DevOps: JUnit, Mockito, Testcontainers, WireMock, Jest, Docker, GitHub Actions, Gradle, Git
Languages: Romanian (native), English (C1)